

1. Setup a highly available ecs cluster with load balancer and dynamic port mapping.

NOTE:

Your cluster should maintain atleast 6 tasks at any point of time and should be highly available across multiple AZ's.

Use the below image from deployment:
sabair0509/hiring-app:works

Goal in simple words:

Run your Docker app on many servers so that

- always **6 containers (tasks) are running**
- if one server dies → app still works
- traffic automatically spreads using **Load Balancer**
- ports auto-assigned (**dynamic port mapping**)
- multi-AZ = high availability

We will use:

- **ECS (EC2 launch type)** → runs containers
- **Auto Scaling Group** → keeps EC2 machines alive
- **Application Load Balancer (ALB)** → traffic distribution
- **Target group (dynamic ports)** → routing to containers
- **Service (desired count = 6)** → keeps 6 tasks always running

Image to deploy:

sabair0509/hiring-app:works

Big Architecture (understand first)

Internet → ALB → Target Group → ECS Service → Tasks → EC2 Instances → Across AZs

If

- one container stops → ECS starts new
- one EC2 dies → ASG launches new
- one AZ fails → other AZ still serving

That = **Highly Available**

Internet

↓

ALB (Port 80)

↓

Target Group (dynamic ports like 32768, 32769...)

↓

ECS Service (6 tasks)

↓

Containers (sabair0509/hiring-app:works → 8080)

↓

EC2 Instances (multiple AZs)

STEP 1 — Create ECS Cluster (EC2 capacity)

Go to:

ECS → Clusters → Create cluster

Cluster config

Setting	Value
Cluster name	hiring-cluster
Provisioning model	On-Demand
Instance type	t3.small (free tier ok)
Number of instances	3
Key pair	create or select
VPC	Default
Subnets	Select ALL subnets (important for HA)
Auto assign public IP	ENABLE

Create cluster ⌚ wait 3–5 min

▼ Infrastructure - *advanced* [Info](#)

Configure the manner of obtaining compute resources that will be used to host your application.

Select a method of obtaining compute capacity

Your cluster is automatically configured for AWS Fargate (serverless), but you may choose to add Amazon EC2 instances (servers).

☐ Fargate only
Serverless - you don't think about creating or managing servers. Great for most common workloads.

☐ Fargate and Managed Instances
Managed instances - Amazon ECS will manage patching and scaling on your behalf while giving you configurability about the types of instances. Great for more advanced workloads.

☒ Fargate and Self-managed instances
Self-managed instances - you must ensure the instances are properly configured and scaled properly, and you have full control over the instances.

Auto Scaling group (ASG) | [Info](#)

☒ Create a new Auto Scaling group - *advanced*

☐ Use an existing Auto Scaling group

Provisioning model

Select a provisioning model for your instances

☒ On-demand
With on-demand instances, you pay for compute capacity by the hour, with no long-term commitments or upfront payments.

☐ Spot
Amazon EC2 Spot instances let you take advantage of unused EC2 capacity in the AWS cloud. [Learn more](#)

Container instance Amazon Machine Image (AMI)

Choose the Amazon ECS-optimized AMI for your instance.

Amazon Linux 2023

EC2 instance type

Choose based on the workloads you plan to run on this cluster.

t3.micro
x86_64
2 vCPU 1 GiB Memory

EC2 instance role

An instance role is used by Amazon EC2 instances to make AWS API requests. If you don't already have an instance IAM role created, we can create one for you.

Create default role

 [Create new instance profile](#)

Desired capacity

Specify the number of instances to launch in your cluster.

Then check:

ECS → Clusters → Infrastructure → **3 instances must show ACTIVE**

Dashboard AWS Global View ↗ Events ▼ Instances Instances Instance Types	Find Instance by attribute or tag (case-sensitive) All states ▾ running X Clear filters						
	☐	Name 🔗 ▾	Instance ID	Instance state ▾	Instance type ▾	Status check	Alarm status
	☐	ECS Instance - ...	i-036e6eb82be58c8b5	Running 🔍 🔍	t3.micro	3/3 checks passed	View alarms +
	☐	ECS Instance - ...	i-04d745556d07c575f	Running 🔍 🔍	t3.micro	3/3 checks passed	View alarms +
	☐	ECS Instance - ...	i-0aff66bfd2921a9af	Running 🔍 🔍	t3.micro	3/3 checks passed	View alarms +

Clusters (1) Info Search clusters By default, we only load up to 1,000 clusters at a time.						
Last updated February 12, 2026, 20:13 (UTC+5:30) 🔄 Create cluster						
< 1 > ⚙️						
Cluster ▾	Services ▾	Tasks ▾	Container instances ▾	CloudWatch monitoring ▾	Capacity provider strategy	
hurring-cluster	1	2 Pe... 5 Ru...	5 EC2	Default	ASG	

STEP 2 — Create Load Balancer

EC2 → Load Balancers → Create → Application Load Balancer

Basic

Field	Value
Name	my-alb
Scheme	Internet facing
IP	IPv4

Network

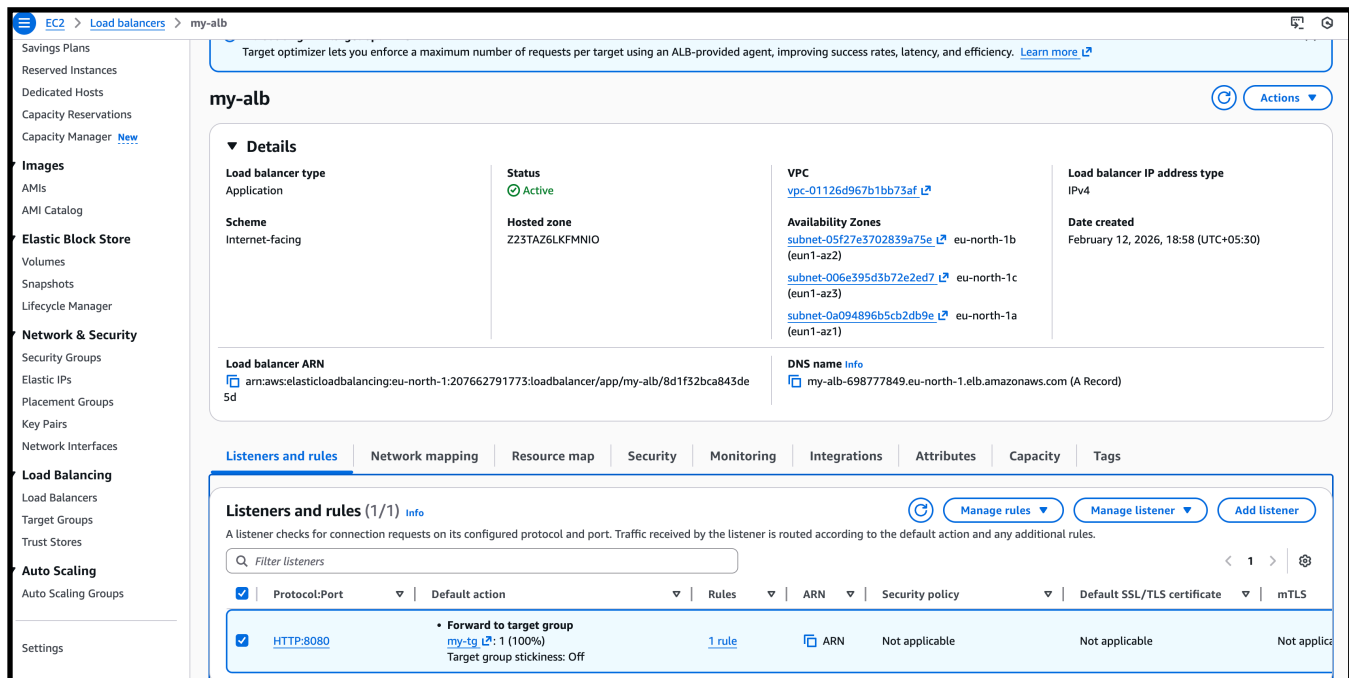
Select **ALL subnets** (very important for multi-AZ HA)

Security group

Create new:

Allow HTTP 80 from Anywhere

Allow Custom TCP 8080 from Anywhere (optional but safe)



STEP 3 — Create Target Group

Field	Value
Type	Instances
Protocol	HTTP
Port	8080
Name	my-tg
VPC	default

Health check

Setting	Value
Path	/hiring/
Success codes	200-499
Port	Traffic port

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
my-tg	arn:aws:elasticloadbalancing:eu-north-1:207662791773:loadbalancer/app/my-alb/8d1f32bca843de5d	8080	HTTP	Instance	my-alb	vpc-01126d967b1bb73af

STEP 4 — Attach TG to ALB

Go to:
Load Balancer → Listeners → Add listener

Create:

Protocol	Port	Forward
HTTP	80	my-tg

Protocol/Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS
HTTP:8080	Forward to target group my-tg (100%) Target group stickiness: Off	1 rule	ARN	Not applicable	Not applicable	Not applicable

STEP 5 — Create Task Definition

ECS → Task Definitions → Create

Launch type

EC2
Network mode = bridge (IMPORTANT for dynamic port mapping)

Task size

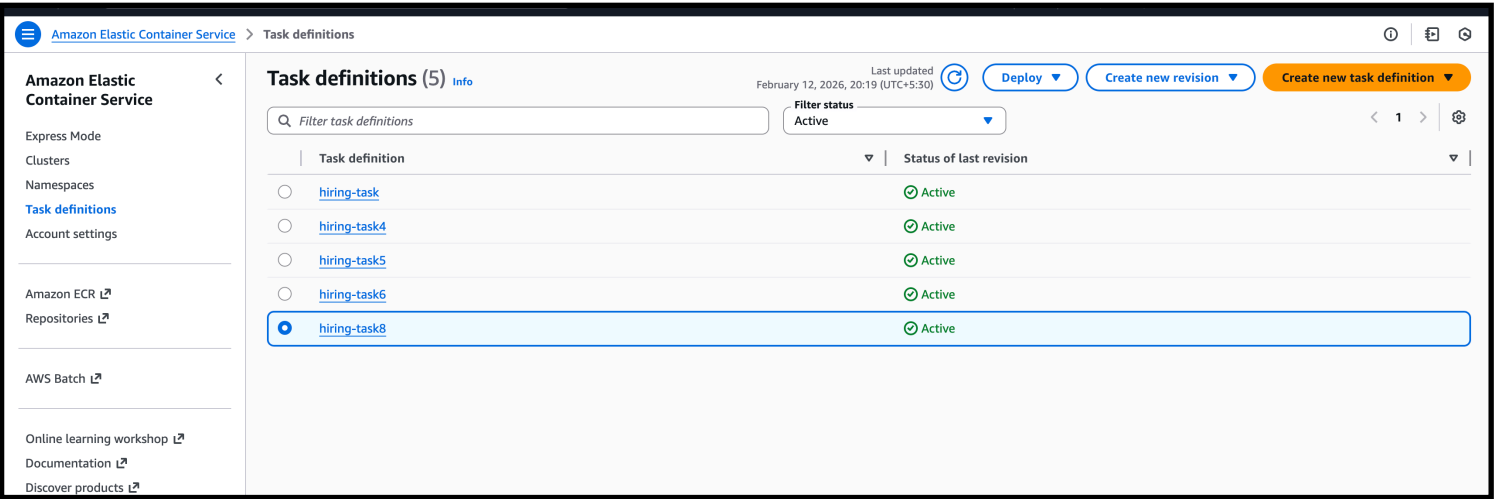
CP U	Memor y
256	512

Container

Field	Value
Name	app
Image	sabair0509/hiring-app:works
Container port	8080
Host port	0 ← ★ DYNAMIC PORT MAPPING

DO NOT add environment variables
DO NOT add volumes

Create

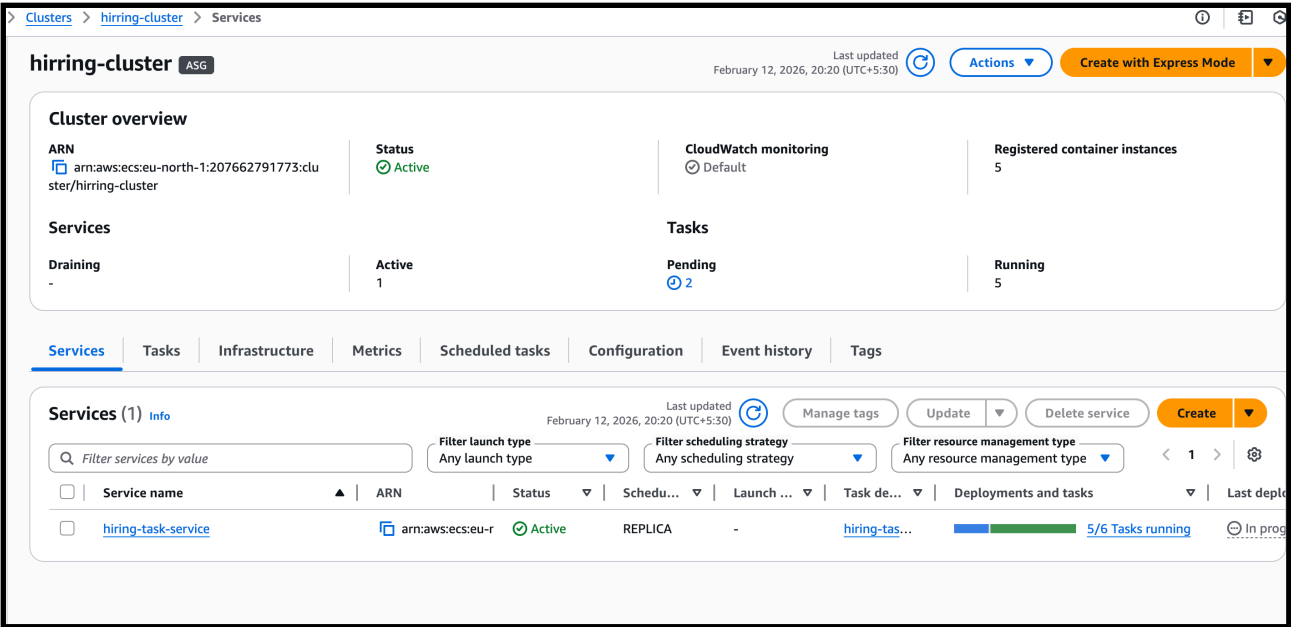


—> I have went through many modification this was the final modification which was success and running

STEP 6 — Create Service

ECS → Cluster → Services → Create

Setting	Value
Launch type	EC2
Task definition	hiring-task
Service name	hiring-service
Desired tasks	6



The screenshot shows the Amazon Elastic Container Service (ECS) console. The breadcrumb navigation at the top indicates the path: Amazon Elastic Container Service > Clusters > hiring-cluster > Services > hiring-task-service > Tasks. The left sidebar contains navigation links for Express Mode, Clusters, Namespaces, Task definitions, Account settings, Amazon ECR, Repositories, AWS Batch, Online learning workshop, Documentation, Discover products, and Subscriptions. The main content area displays a table of tasks for the 'hiring-task-service'.

Tasks (1/20)

Filter tasks by property or value: Filter desired status: Any desired status Filter launch type: Any launch type

Task	Last status	Desired status	Task definition	Health status	Created at	Started by	Size
☒ 2dffa92c2fad4ceba990fb...	Running	Running	hiring-task6:1	Unknown	34 minutes ago	ecs-svc/13115784301...	34
☐ 3a0de915b7924001be70...	Running	Running	hiring-task6:1	Unknown	33 minutes ago	ecs-svc/13115784301...	33
☐ 61346f593311476b90ab...	Running	Running	hiring-task6:1	Unknown	33 minutes ago	ecs-svc/13115784301...	33
☐ c580c9543e204b6091b5...	Running	Running	hiring-task6:1	Unknown	33 minutes ago	ecs-svc/13115784301...	31
☐ ce8dbad08b6247e7950a...	Running	Running	hiring-task6:1	Unknown	33 minutes ago	ecs-svc/13115784301...	31
☐ a39e347ed8a34d5a8fe58...	Provisioning	Running	hiring-task2	Unknown	20 minutes ago	ecs-svc/14539380324...	-

Containers for task 2dffa92c2fad4ceba990fbf49a3f79e

Containers (1)

Filter containers:

Container name	Container runtime ID	Image URI	Image Digest	Status	Health status	CPU	Memory hard/soft limit
app	ce3fd963d65442...	sabair050...	sha256:5...	Running	Unknown	0	- / -

—> container is running

Placement

Spread across AZ

Spread across Instance

Create service ⌚

STEP 6 — Verify

—> check your any server running and

—> docker images

—> docker ps

```
waseemakram@waseem Downloads % ssh -i "ansible.pem" ec2-user@ec2-16-90-58.eu-north-1.compute.amazonaws.com
#
#####
~\
~\#####\
~\#####\
~\#/\
~\V~' '-> Amazon Linux 2023 (ECS Optimized)
~\
~\
~\
~\m/'

For documentation, visit http://aws.amazon.com/documentation/ecs
Last login: Thu Feb 12 12:41:08 2026 from 14.192.14.54
[ec2-user@ip-172-31-8-177 ~]$ sudo -i
[root@ip-172-31-8-177 ~]# docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
amazon/amazon-ecs-agent   latest             2cd16d1f9fa1       12 days ago        120MB
ebs-csi-driver          latest             b479ea6bce25       12 days ago        58.8MB
ecs-service-connect-agent interface-v1        369a1f77771a       6 weeks ago        204MB
sabair0509/hiring-app    works             b49fe6b94ba1       2 years ago        426MB
amazon/amazon-ecs-pause  0.1.0             9dd4685d3644       11 years ago       702kB
[root@ip-172-31-8-177 ~]# docker ps
CONTAINER ID   IMAGE                                COMMAND              CREATED             STATUS              PORTS                               NAMES
ce3fd963d654   sabair0509/hiring-app:works        "catalina.sh run"    36 minutes ago      Up 36 minutes      0.0.0.0:32768->8080/tcp, :::32768->8080/tcp   ecs-hiring-task
-1-app-80d7cc88efb2f2a3d501
c5712a47ab90   amazon/amazon-ecs-agent:latest     "/agent"             2 hours ago         Up 2 hours (healthy)                               ecs-agent
[root@ip-172-31-8-177 ~]#
```

—> now check your port 32768/hiring



STEP 7 — Final URL

Open: load balancer url

`http://my-alb-xxxx.elb.amazonaws.com:8080/hiring/`



RESULT (Assignment completed)

You now have:

- ✓ Highly available (multi-AZ)
- ✓ 6 running tasks always
- ✓ Auto placement across instances
- ✓ Dynamic port mapping
- ✓ Load balancer routing traffic

